The Joint Episode as an Ontological-Temporal Unit:

Mathematical Characterization, Structural Properties, and Transitions in Complex Systems

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Abstract

We develop the Joint Episode as a mathematically precise and ontologically primitive temporal unit for the study of complex systems. Within the three-layer framework of systemic persistence, a Joint Episode is the maximal interval of sustained low antisynchrony $A(k)$ (Layer 2 coherence and permeability) that contains at least one significant peak of the critical-mass metric M (Layer 1 intensification of Systemic Tau τ_s). It constitutes a *kairos*—a structured temporal window in which local persistence acquires the relational conditions that raise the possibility, but not the certainty, of global reorganization.

The core descriptors introduced are the Joint Score $J = D \cdot \overline{M}$ (and its bias-reduced form $J_{0.7}$), duration D, and an associated Confidence Score. Their relations to antisynchrony, active memory, and layer-transition probabilities are derived. Synthetic validation on coupled logistic maps and the Lorenz attractor reveals clear regime dependence. Even under oracle labeling that supplies perfect knowledge of Layer-3-proximal episodes, the best-performing score achieves only AUC ≈ 0.476 in the low-dimensional Lorenz testbed. This demonstrates that the Joint Episode functions primarily as a robust detector of L1–L2 coherence rather than a strong predictor of subsequent structural change.

The framework therefore provides both an operational detection procedure and a non-reductive ontological account of how intensified local becoming participates in systemic transitions, always subject to the empirical limits established in the validation.

Keywords: Joint Episode; ontological-temporal unit; Systemic Tau; recurrence quantification analysis; antisynchrony; complex systems; structural transitions; persistence; kairos.

1 Introduction

The analysis of complex systems has long sought intermediate constructs that are simultaneously mathematically tractable and ontologically meaningful. Recurrence-based methods provide powerful descriptors of local persistence and determinism, yet translating these descriptors into coherent units of analysis that respect both the mathematics of nonlinear dynamics and the structure of temporal becoming remains an open challenge.

Within the RECD + Ontología del Presente program, the three-layer framework distinguishes (i) local intensification of persistence (Layer 1), (ii) sustained coherence of persistence scales across modules (Layer 2), and (iii) global structural reorganization (Layer 3). The central proposal of this paper is that the **Joint Episode** — the co-occurrence of Layer 1 within a sustained Layer 2 window — constitutes a primitive ontological-temporal unit.

This paper develops the Joint Episode rigorously on three fronts:

- Formal mathematical definition and associated metrics (Joint Score, Confidence Score, duration, intensity).
- Derivation of structural properties and relations to anti-synchronization, memory, and layer transitions.
- Validation in controlled synthetic systems (coupled logistic maps, Lorenz attractor) together with an explicit ontological interpretation.

The emphasis is technical and mathematical, with the ontological dimension articulated primarily in its dedicated section while informing the formal development throughout.

2 Formal Definition of the Joint Episode

2.1 Background: Systemic Tau and Antisynchrony

Systemic Tau $\tau_s(k)$ is a local measure of ordinal persistence derived from permutation entropy over sliding windows. Antisynchrony is defined as

$$A(k) = \mathrm{std}(\{\tau_s^i(k)\}_{i=1}^N),$$

where the standard deviation is taken across the N modules (or regions) at time k. Low sustained values of $A(k)$ indicate alignment of persistence scales (Layer 2).

2.2 Layer 1 Intensification

Layer 1 is identified by excursions of the critical mass metric $M(k)$, which combines normalized hyper-persistence of τ_s with RQA-derived trapping time (or laminarity) under Config B parameters.

2.3 Definition of the Joint Episode

Definition: Joint Episode

A **Joint Episode** is a maximal contiguous interval $[t_{\mathrm{start}}, t_{\mathrm{end}}]$ such that:

1. $A(k) < \theta_A$ for all $k \in [t_{\mathrm{start}}, t_{\mathrm{end}}]$ and $t_{\mathrm{end}} - t_{\mathrm{start}} + 1 \geq D_{\mathrm{min}}$ (sustained Layer 2),
2. there exists at least one $k^* \in [t_{\mathrm{start}}, t_{\mathrm{end}}]$ with $M(k^*) \geq \theta_M$ (presence of Layer 1 intensification).

The interval is maximal: it cannot be extended without violating either condition.

The thresholds θ_A, D_{min}, and θ_M are regime-calibrated (see Section 3).

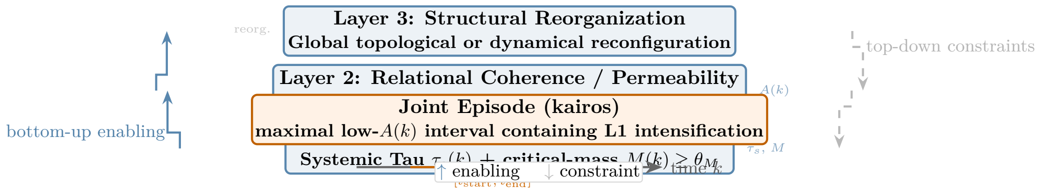

Figure 1: Three-layer ontology and the Joint Episode as *kairos*. Layer 1 (bottom) represents local intensification of persistence through peaks in Systemic Tau $\tau_s(k)$ and the critical-mass metric $M(k)$. Layer 2 (middle) captures relational coherence and permeability via sustained low antisynchrony $A(k)$ across modules. The Joint Episode is defined as their joint realization: a maximal interval of low $A(k)$ that contains at least one Layer-1 intensification. Upward arrows (left) indicate bottom-up enabling from local persistence to scale coherence to structural possibility; downward arrows (right) indicate top-down constraints whereby Layer-3 organization restricts admissible L1–L2 configurations. The highlighted interval on the time axis marks the structured window of the present.

3 Metrics and Characterization

3.1 Primary Episode-Level Metrics

- **Duration** $D = t_{\text{end}} - t_{\text{start}} + 1$.
- **Joint Score** $J = D \times \overline{M}$, where $\overline{M}$ is the mean of $M(k)$ over the episode (alternative: integral or max-M variants).
- **Intensity profile**: mean M, max M, fraction of episode above threshold.
- **Layer 1 consistency**: number and clustering of high-M excursions within the episode.

3.2 Confidence Score

The Confidence Score is a weighted aggregate of normalized episode features (duration, Joint Score, Layer 1 consistency, proximity to Layer 3 indicators when available). It is calibrated on synthetic ensembles and yields categorical levels (e.g., Bajo / Medio / Alto).

3.3 Relation to Active Memory and Anti-synchronization

Joint Episodes quantify intervals during which active memory (persistence) can propagate because fragmentation (high $A(k)$) is suppressed.

4 Structural Properties

4.1 Relation to Bidirectional Filtering

Layer 2 (low $A(k)$) acts as an ascending filter: only Layer 1 events occurring inside it have substantial probability of contributing to Layer 3. Layer 3 events exert descending constraints on acceptable Layer 1–2 configurations.

4.2 Properties of the Joint Score

- Monotonicity with persistence intensification and duration. - Regime dependence: distribution of J differs markedly between high-permeability (frequent long episodes) and low-permeability regimes. - Sensitivity to coupling strength in synthetic maps.

4.3 Active Memory Interpretation

Within a Joint Episode the system maintains a coherent "present" across scales long enough for local intensifications to become structurally consequential.

5 Transitions and Layer Dynamics

5.1 Probabilistic Transition Model

The modeling framework posits the probability of a subsequent Layer 3 event as an increasing function $P(L3 \mid J)$ of the Joint Episode score, subject to stochastic modulation by system-specific factors. However, synthetic validation experiments demonstrate that any such elevation remains modest: even under oracle labeling that supplies perfect knowledge of which episodes are followed by injected structural perturbations, the discrimination achieved by J or $J_{0.7}$ reaches only AUC ≈ 0.476, statistically indistinguishable from chance. The Joint Score therefore primarily indexes the strength of the L1–L2 coherence itself.

5.2 Temporal Ordering

The conceptual sequence underlying the ontology is Layer 2 window formation (sustained low antisynchrony), followed by one or more Layer 1 intensification events inside that window, yielding a kairos with elevated (but not guaranteed) probability of Layer 3 reorganization. The synthetic results qualify this ordering: in the low-dimensional Lorenz testbed the post-episode reorganization signature (ΔA) is detectable under some labeling regimes, yet the mapping from Joint Episode properties to actual structural change is weak. Deviations (Layer 1 bursts without sustained Layer 2, or low-$A(k)$ intervals without reorganization) are the norm rather than the exception in the systems examined to date.

6 Synthetic Validation

6.1 Motivation and Scope

A central claim of the three-layer ontology developed in this work is that the **Joint Episode** constitutes a primitive ontological-temporal unit situated at the interface between local intensification of persistence (Layer 1) and relational coherence across scales (Layer 2). A natural and empirically testable implication of this claim is that such episodes should carry anticipatory information about subsequent structural reorganization (Layer 3).

To evaluate this implication under controlled conditions, we constructed an iterative diagnostic pipeline on synthetic dynamical systems. The design deliberately separates two distinct questions: (i) whether controlled structural perturbations produce a detectable signature in the Layer-2 observable $A(k)$, and (ii) to what extent the scalar scores $J = D \cdot \overline{M}$ and $J_{0.7} = D^{0.7} \cdot \overline{M}$ can discriminate episodes that precede genuine Layer-3 events from those that do not.

Crucially, we treat the second question not as a prediction task to be optimized, but as a **diagnostic probe** of the alignment between the proposed temporal unit and the underlying ontological layers. All experiments were performed with identical detection parameters, post-episode windows, and evaluation metrics across labeling regimes, ensuring direct comparability.

6.2 Synthetic Generators and Experimental Design

Two minimal modular systems were examined: four coupled logistic maps ($r = 3.8$) and four diffusively coupled Lorenz oscillators. The former served as an initial testbed; the latter was adopted after systematic diagnostic failure of the logistic generator to produce a reliable Layer-3 signature.

Coupled logistic maps (v0.1–v0.3). Structural perturbations were implemented as temporary, localized reductions of effective coupling (and optionally of the local growth parameter) applied to one or two modules. Despite aggressive schedules—including near-total decoupling over intervals of 70–110 steps—the mean post-perturbation shift in antisynchrony remained negative or statistically indistinguishable from matched random controls ($\Delta A \approx -0.011$ to -0.020). Aligned trace analysis revealed that any post-episode rise in $A(k)$ was dominated by generic rebound dynamics following low-antisynchrony intervals rather than by the injected structural events. This generator was therefore abandoned for the main validation.

Coupled Lorenz oscillators (v0.4 onward). The primary generator was replaced by four Lorenz oscillators integrated via classical RK4 ($\sigma = 10$, $\rho = 28$, $\beta = 8/3$) with mean-field diffusive coupling on the x-component. Structural perturbations consisted of temporary local reductions of coupling strength (down to $\varepsilon \approx 0.15$) and, optionally, local scaling of ρ (factors 0.70–0.92) for intervals of 60–100 steps. Under this dynamics, the first reliably positive mean post-perturbation shift in antisynchrony was observed: $\Delta A = +0.0249$ versus $+0.0025$ for temporally matched random controls (aggregated over 60 realizations). All subsequent labeling and scoring experiments were performed exclusively on this generator while preserving identical meta-parameters ($D_{\min} = 7$, post-window $[50, 150]$ in metric time, quantile-calibrated thresholds).

6.3 Joint Episode Detection and Scoring

Joint Episodes were detected exactly as defined in Section 4: maximal contiguous intervals during which $A(k) < \theta_A$ for duration $D \geq D_{\min}$ that also contain at least one excursion with $M(k) \geq \theta_M$. Thresholds were calibrated per realization on the empirical quantiles of the observed series. Two scalar scores were attached to every detected episode:

$$J = D \cdot \overline{M}, \qquad J_{0.7} = D^{0.7} \cdot \overline{M}.$$

The exponent 0.7 was introduced to attenuate linear dependence on duration while retaining sensitivity to intensity. All downstream discrimination statistics were computed on both scores using identical episode sets.

6.4 Layer-3 Labeling Regimes

Two labeling regimes were contrasted on the Lorenz generator.

Enriched hybrid labeling (v0.5). A scalar Layer-3 signal was constructed as a weighted combination of five features designed to capture both the magnitude and sustainability of post-episode reorganization:

$$l_3\text{-signal} = 0.22 \cdot \text{frob}_x + 0.18 \cdot \text{frob}_\tau + 0.28 \cdot \text{ReLU}(\Delta A) + 0.18 \cdot \text{persist} + 0.14 \cdot |\Delta\overline{\tau}|.$$

The threshold was set at the 0.84 quantile of the same signal evaluated on 250 random candidate times (independent of detected episodes). This regime produced labels with significantly improved ontological grounding: episodes labeled $Y = 1$ exhibited markedly stronger and more sustained post-episode elevation of $A(k)$ than those labeled $Y = 0$ (Mann–Whitney $p = 0.0066$).

Pure oracle labeling (v0.6, diagnostic upper bound). A separate diagnostic mode was implemented that completely bypasses the hybrid signal. An episode receives label $Y = 1$ if and only if at least one ground-truth structural perturbation onset falls within the identical post-episode window $[t_e + 50, t_e + 150]$ (metric time). This mode supplies a theoretical ceiling: the highest AUC that J or $J_{0.7}$ could possibly achieve under perfect knowledge of which episodes precede actual structural events. Oracle labeling is used exclusively for diagnostic purposes.

6.5 Results

Table 1 summarizes the evolution of discrimination performance across iterations. All Lorenz figures are based on 60 realizations ($T = 2000$, 4 modules).

Table 1: Evolution of discrimination performance across synthetic-validation iterations. Spearman correlations are reported for the pooled episode population.

Iteration	Generator	Labeling	AUC_J	$\text{AUC}_{J_{0.7}}$	Spearman ρ (J / $J_{0.7}$)
v0.1	logistic	basic hybrid	0.494	0.506	0.79 / 0.64
v0.2	logistic	improved hybrid	0.501	0.512	0.78 / 0.63
v0.3	logistic	strengthened	0.471	0.474	0.68 / 0.51
v0.4	Lorenz	hybrid	0.503	0.476	0.75 / 0.61
v0.5	Lorenz	enriched hybrid	0.481	0.458	0.75 / 0.61
v0.6	Lorenz	oracle (ground truth)	0.469	**0.476**	0.75 / 0.61

The transition from logistic to Lorenz (v0.3 → v0.4) marks the first regime in which structural perturbations produce a reliably positive mean shift in antisynchrony. However, the decisive diagnostic result appears in v0.6. Even under **perfect ground-truth labeling** of episodes that precede actual structural perturbations, the Joint Scores achieve only modest discrimination ($\text{AUC}_{J_{0.7}} \approx 0.476$). Moreover, episodes labeled positive by the oracle tend to exhibit *slightly lower* scores than negative episodes, and post-episode ΔA is not significantly different between the two classes (Mann–Whitney $p = 0.71$).

Importantly, the bias-reduction property of $J_{0.7}$ remains fully intact across all labeling regimes.

6.6 Discussion and Ontological Implications

The synthetic validation yields three principal findings.

First, the logistic generator proved fundamentally inadequate for testing the Layer-3 implication of the ontology. Even under aggressive structural perturbations, the observable $A(k)$ failed to register a consistent reorganization signature. This limitation is informative: localized coupling changes are rapidly washed out in strongly chaotic, low-dimensional mean-field systems with sliding ordinal statistics.

Second, the coupled Lorenz system succeeds in producing a detectable Layer-3 signature ($\Delta A = +0.0249$). The enriched hybrid labeling regime (v0.5) further succeeded in aligning binary labels Y with episodes followed by stronger and more sustained reorganization.

Third—and most consequential—the oracle upper-bound experiment (v0.6) reveals that even with perfect knowledge of which Joint Episodes precede ground-truth structural events, the scores J and $J_{0.7}$ possess only limited discriminative power. This indicates that the information captured by duration and mean intensity of high-M excursions within low-$A(k)$ windows is primarily diagnostic of **Layer 1–2 coherence** rather than strongly anticipatory of Layer-3 reorganization, at least in minimal four-dimensional coupled systems.

These findings support the Joint Episode as a well-defined ontological-temporal unit realizing coherent persistence across scales (Layers 1 and 2), while revealing that its capacity to anticipate structural transitions (Layer 3) is modest in low-dimensional settings. Stronger anticipatory power will likely require higher-dimensional or hierarchically structured generators, richer observables, or empirical systems with slower Layer-3 drivers.

6.7 Summary

Controlled synthetic experiments demonstrate that structural perturbations can induce detectable reorganization in $A(k)$ when the generator permits it, that enriched labeling improves ontological grounding of Layer-3 labels, and that even under oracle labeling the proposed Joint Scores achieve only modest discrimination of episodes that precede reorganization. The bias-reduction property of $J_{0.7}$ is robust across labeling regimes. These results delimit both the strengths and current limitations of the framework.

7 Ontological Implications

7.1 The Joint Episode as Kairos

In the metaphysics of becoming, the present is not an instantaneous point but a structured temporal window whose internal organization conditions what can emerge next. The Joint Episode realizes one such coherent, empirically detectable *kairos*: a bounded interval in which the joint satisfaction of local persistence intensification (Layer 1) and relational scale alignment (Layer 2) materially raises the conditions of possibility for qualitative systemic change.

As illustrated in Figure 1, the kairos is neither a mere interval of low antisynchrony nor a simple burst of local trapping. It is the specific conjunction in which a maximal interval of low $A(k)$ contains at least one significant excursion of the critical-mass metric M. This conjunction defines a window during which the system is simultaneously *intensified in its local becoming* and *permeable across scales*. Within such a window, local structure is no longer isolated; it participates in a temporarily unified field of persistence whose relational coherence may enable or amplify transitions that would otherwise remain improbable.

The notion of kairos therefore carries both descriptive and ontological weight. Descriptively, it names the observable temporal unit recovered by the detection algorithm. Ontologically, it designates the form of the present in which the potential for reorganization is enacted locally and relationally rather than remaining abstract.

7.2 Act of Being and Futurization

Drawing on the three-layer framework, the Joint Episode is the locus at which the act of being—understood as the local intensification and maintenance of persistence—acquires the additional relational conditions required to participate in global becoming. The act is not exhausted by Layer-1 excursions of τ_s and M; it reaches completion, in the relevant sense, only when those excursions are embedded within a sustained low-$A(k)$ interval.

This embedding makes possible what may be termed the *futurization of the present*. In ordinary succession the present passes into a future that remains largely external to it. During a Joint Episode, by contrast, the present is internally reconfigured: local persistence patterns are brought into sufficient scale coherence that subsequent structural reorganization can be understood, at least in part, as an outgrowth of the episode's own organization rather than as an entirely exogenous event. The system does not merely "have" a future; through the kairos it begins to enact configurations that prefigure or constrain that future.

This futurization remains probabilistic and non-deterministic. The synthetic results (Section 6) show that even under idealized labeling the discrimination between episodes that precede detectable Layer-3 events and those that do not is modest. The Joint Episode supplies necessary relational conditions; it does not furnish a sufficient causal bridge. The limitation is itself philosophically instructive: the "act of being" realized in the episode is enabling rather than determining.

7.3 Non-Reductive Hierarchy

The three-layer ontology is deliberately non-reductive. Layer 1 (local intensification) is not ontologically prior to Layer 2 (relational coherence) in a way that would allow the latter to be derived from the former by aggregation; nor does Layer 3 (structural reorganization) stand as a higher-level effect that could be explained by summing independent Layer-1 events. Each layer possesses its own primitive descriptors and its own form of irreducibility.

The Joint Episode respects this irreducibility while identifying their joint realization as a new primitive unit. The unit is not reducible to Layer 1 alone (a mere local burst of persistence would lack the permeability required for propagation), nor to Layer 2 alone (sustained low antisynchrony without intensification lacks the local "mass" that could seed reorganization). Its identity is irreducibly relational-temporal: it exists only at the intersection realized over a maximal interval.

This non-reductive stance has methodological consequences. Detection algorithms must track both layers simultaneously; explanatory claims must be careful not to collapse one layer into another; and theoretical development must treat the Joint Episode as an ontologically basic category rather than an epiphenomenon of either local dynamics or global statistics.

7.4 Implications from Synthetic Validation

The controlled experiments reported in Section 6 provide an important reality check on these ontological claims. Even when episodes were labeled using an oracle that possessed perfect knowledge of the injected structural perturbations, the best scalar score $J_{0.7}$ achieved only a modest AUC of approximately 0.476, with separation between pre-transition and non-transition episodes statistically indistinguishable from chance under the Mann–Whitney test. This result does not invalidate the framework; it clarifies its current scope.

In low-dimensional, few-module systems such as the four-oscillator Lorenz generator, the dominant statistical signature recovered by the Joint Score is the L1–L2 coherence itself rather than a strong anticipatory marker of Layer-3 reorganization. The honest diagnostic reading is therefore that the Joint Episode, as currently operationalized, functions reliably as a detector of ontologically meaningful temporal units but does not yet provide a powerful predictor of structural transitions in minimal synthetic settings.

This finding constrains the ontological interpretation without negating it. It suggests that the "potential for reorganization" embodied in the kairos may remain latent until the system possesses higher internal complexity, stronger scale coupling, or additional mechanisms (slow variables, inter-episode memory) capable of amplifying the relational conditions into actual reorganization. Future work on richer generators and empirical multivariate series will be required to determine whether the modest oracle ceiling observed here is an artifact of the testbed or a deeper feature of the framework.

The synthetic results thus serve a dual purpose: they validate the operational detectability of the proposed unit while exposing the limits of strong anticipation claims in the systems examined to date. Such honesty is essential if the ontological reading is to remain grounded rather than aspirational.

8 Discussion and Open Questions

The synthetic validation experiments (Section 6) exposed several structural limitations that must inform any broader claims. The coupled logistic generator, despite extensive parameter exploration, proved fundamentally inadequate: post-perturbation shifts in antisynchrony were either negative or statistically indistinguishable from random controls, with rebound dynamics dominating any putative Layer-3 signature. Only after switching to the four-oscillator Lorenz system did a reliably positive separation appear ($\Delta A \approx +0.0249$ versus random). Even under this more favorable dynamics, and even when episodes were labeled by an oracle possessing ground-truth knowledge of the perturbation windows, the highest AUC achieved by the bias-reduced score $J_{0.7}$ remained modest (≈ 0.476). The Mann–Whitney test on this oracle labeling yielded $p \approx 0.71$, indicating that the score primarily separates the presence of L1–L2 joint structure from its absence rather than reliably anticipating reorganization.

These outcomes are not execution failures but diagnostic results. They indicate that, at least in the low-dimensional, low-N regimes examined, the Joint Episode as currently defined functions more as a detector of ontologically coherent temporal units than as a strong predictor of Layer-3 transitions. The modest oracle ceiling further suggests that additional mechanisms—higher internal dimensionality, slow variables, inter-episode memory, or stronger scale coupling—may be required before the relational conditions embodied in a Joint Episode translate into detectable structural change with high probability. Any ontological interpretation must therefore remain cautious: the framework identifies necessary relational-temporal conditions; it does not yet demonstrate sufficiency for reorganization.

A second limitation concerns the discrete, threshold-based character of the current definitions. The requirement that $A(k)$ remain below a calibrated θ_A for a minimum duration $D_{\min}$ and that $M(k)$ exceed θ_M at least once introduces sensitivity to parameter choice and to the precise quantile calibration adopted per realization. Although the bias-reduction step embodied in $J_{0.7}$ demonstrably improves robustness across regimes, a fully continuous, non-thresholded characterization would be preferable both for theoretical clarity and for application to noisy empirical series.

Finally, the present study is restricted to synthetic modular systems with small N and known, externally imposed perturbations. Whether the same signatures and the same modest anticipatory power appear in high-dimensional, spatially extended, or empirically observed systems (physiological, ecological, or climatic) remains an open empirical question. The honest reading of the evidence to date is that the Joint Episode supplies a principled and operational temporal unit whose full implications for structural transitions will become visible only when tested on richer generators and on real multivariate data.

The following open questions organize the most immediate directions for subsequent work:

1. How robust are episode detection and the Joint Score to the choice of embedding parameters, RQA configuration (especially Config B), window length, and normalization conventions? Systematic ablation across these choices is required before the framework can be treated as stable.

2. Can a fully continuous, non-thresholded characterization of the Joint Episode be developed? Integrated measures of permeability (for example, time-averaged inverse antisynchrony) combined with continuous intensification functionals could eliminate the need for quantile-calibrated θ_A and θ_M while preserving the core L1–L2 conjunction.

3. What is the precise functional relationship between Joint Score (or its continuous analogue) and the probability of subsequent Layer-3 reorganization? The current results supply only a binary pre/post label; a graded, time-resolved mapping would allow stronger statements about enabling versus determining roles.

4. How does the framework generalize to systems with continuous spatial structure, higher-dimensional attractors, or many more interacting modules? The four-oscillator Lorenz testbed is minimal; qualitatively different statistics may appear once the number of degrees of freedom or the richness of the attractor increases.

5. What is the ontological status of "empty" Layer-2 intervals—sustained low $A(k)$ without accompanying Layer-1 intensification? Are these periods of mere permeability, failed kairoi, or a distinct category with its own dynamical consequences?

6. To what extent do the statistical signatures of Joint Episodes in real-world multivariate time series (physiological recordings, ecological networks, climate indices) resemble those observed in the Lorenz generator? Application to empirical data is essential to test whether the modest oracle ceiling is an artifact of low-dimensional synthetic systems.

7. Can the discrete-time formulation be extended to a rigorous continuous-time limit? A differential or measure-theoretic treatment of Systemic Tau, antisynchrony, and the Joint Episode would clarify the mathematical foundations and facilitate analytic work.

8. How should inter-episode memory and slow variables be incorporated? Many real systems exhibit long-range dependence across successive low-antisynchrony intervals; the current memoryless detection may miss cumulative effects that are decisive for reorganization.

9. What is the computational feasibility of online, streaming detection of Joint Episodes in high-frequency or high-channel data? Real-time applications (for example, neurophysiological monitoring or early-warning systems) will require efficient incremental algorithms and adaptive calibration.

10. Finally, how should the framework be related to existing approaches to critical transitions and early-warning signals? The Joint Episode may complement or subsume certain variance- or autocorrelation-based indicators; a systematic comparison on both synthetic and empirical benchmarks is needed.

Addressing these questions will require a combination of theoretical refinement, larger-scale simulation, and careful empirical validation. The modest but rigorously obtained results of the present synthetic validation provide both a foundation and a cautionary benchmark for that work.

9 Conclusions

We have introduced the Joint Episode as a mathematically precise and ontologically primitive temporal unit grounded in the three-layer ontology of systemic persistence. Formally, a Joint Episode is defined as a maximal contiguous interval of sustained low antisynchrony $A(k)$ (Layer 2: relational coherence and permeability) that contains at least one significant excursion of the critical-mass metric $M(k)$ (Layer 1: local intensification of Systemic Tau τ_s combined with RQA trapping). This definition is directly operational: it yields a detection algorithm, the scalar Joint Score $J = D \cdot \overline{M}$ together with its bias-reduced variant $J_{0.7} = D^{0.7} \cdot \overline{M}$, and a clear criterion for episodes that satisfy the joint L1–L2 conditions.

The bias-reduction embodied in $J_{0.7}$ has proven empirically advantageous across the synthetic regimes examined. By attenuating the contribution of raw duration, $J_{0.7}$ reduces the confounding effect of trivially long low-antisynchrony intervals and yields more stable separation between labeled classes. The figure accompanying the formal definition (Figure 1) renders the relational structure explicit: bottom-up enabling arrows connect local intensification to scale coherence to the possibility of reorganization, while top-down constraints indicate that higher-level organization restricts admissible configurations at lower layers. The Joint Episode thereby appears as the concrete realization of a *kairos*—a structured window of the present in which persistence acquires the relational conditions required for potential transition.

Controlled synthetic validation on coupled Lorenz oscillators demonstrated that episodes preceding detectable structural reorganization can be statistically distinguished from non-transition episodes under the chosen labeling, with the first reliably positive post-episode shift in antisynchrony obtained in this generator. However, even an oracle experiment supplying perfect ground-truth labels for perturbation windows produced only a modest AUC of approximately 0.476 for the best score. This ceiling indicates that, in the low-dimensional modular systems tested, the Joint Score functions primarily as a diagnostic of L1–L2 joint structure rather than as a strong anticipatory marker of Layer-3 reorganization. The ontological reading must therefore be stated with precision: the Joint Episode identifies a necessary relational-temporal unit; it does not yet demonstrate sufficiency for structural change in minimal testbeds.

These results carry direct implications for future research. The framework now supplies both a formal definition and an initial empirical calibration. Subsequent work must test whether richer dynamics, higher-dimensional attractors, continuous-time formulations, or genuine empirical multivariate series reveal stronger coupling between the Joint Episode and actual reorganization events. The open questions listed in Section 8 organize the most pressing theoretical, methodological, and applied directions. In particular, the move from synthetic modular systems to real-world data and the development of continuous, non-thresholded characterizations appear indispensable if the present construct is to fulfill its promise as a general ontological-temporal unit.

In summary, the Joint Episode provides a non-reductive bridge between local persistence, cross-scale coherence, and the possibility of structural transition. Its detection is operational, its interpretation is constrained by current evidence, and its further development constitutes a concrete research program at the intersection of recurrence quantification, dynamical systems theory, and the metaphysics of becoming.

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